Quantum Measurement, Gravitation, and Locality in the Dynamic Theory

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Abstract

This presentation gives an overview of the Dynamic Theory which offers classical thermodynamics as a new basis for the various physical theories. This leads to a five dimensional description of nature where the five dimensions are space, time, and mass. Just as in thermodynamics where an integrating factor, the temperature, ties the energy, heat, to the entropy, the Dynamic Theory produces two metrics tied by a geometrical integrating factor. One metric is integrable and, and necessarily Riemannian, while the other is a non-integrable Weyl space. In the geometrical picture entropy is the arc length of the Riemannian space. The gauge function which appears in the Weyl space leads to a ten-component gauge field tensor in which the gravitational vector field and the gravitational potential represent four components with the electric and magnetic vector fields representing the remaining six components. Isentropic states lead to a null trajectory in the entropy space, which is Einstein?s trajectory of light, but lead to an infinite number of null trajectories in the Weyl space which are given be a generalization of quantum mechanics. The null trajectories in the Weyl space, which are quantum states, give rise to quantization of the gauge field in terms of integer units of charge and lead to a non-singular gauge potential of the form $\frac{1}{r}\, exp[-(\frac{\lambda}{r})]$. The fact that the gravitational field appears as a gauge field in the five-dimensional field tensor, which comes from the gauge function, means that the gravitational field influences the quantum measurement (i.e., the least unit of action) and this shows up in the prediction of the red shifts from stellar objects. Since the gravitational potential is a non-singular form the predicted advance of the perihelion of binary stars is half that of Einstein?s General Relativity which makes a better agreement with experiment, yet yields that same prediction as General Relativity when one body is very massive compared to the other.

I. INTRODUCTION

Einstein used his postulate concerning the constancy of the speed of light in showing that there was a limiting velocity for material objects. This limiting velocity is similar to the limiting aspect of the absolute zero temperature that appears in classical thermodynamics. This gives rise to the question as to whether or not there might be a fundamental connection between the two limiting concepts, one in mechanics and the other in thermodynamics. This question was first addressed by the author in 1976[1] and provides the basis for later work[2], yet the author later learned that he was not the only one to have considered such a question[3]. The presentation below will show that the laws of thermodynamics not only require Einstein's postulate concerning the speed of light, but also require that gravitation, locality, and quantum measurement are interrelated in a very fundamental manner.

II. FIRST LAW (CONSERVATION OF ENERGY)

The concept of conservation of energy is fundamental to all branches of physics and is the beginning of thermodynamics and mechanics[1, 2]. In terms of generalized coordinates or independent variables, the notion of work, or mechanical energy, is considered linear forms of the type

$$\bar{d}w = F_i(q^1, \ldots, q^n, u^1, \ldots, u^n)\; dq^1 \quad (i = 1, 2, \ldots, n). \tag{1}$$

where the forces F_i may be functions of the velocities ($\frac{dq^i}{dt} = u^i$) as well as the coordinates q^i and the summation convention is used.

A system may acquire energy by other means in addition to the work terms; such energy acquisition is denoted dE. The system energy, which represents the energy possessed by the system, is considered to be

$$U(q^1, \ldots, q^n, u^1, \ldots, u^n). \tag{2}$$

With these concepts, the First Law, which is the generalized Law of Conservation of Energy, has the form

$$\begin{aligned} \bar{d}E &= dU - \bar{d}W \\ &= dU - F_i dq^i \;\; (i = 1, \ldots, n). \end{aligned} \tag{3}$$

In the First Law the dimensionality is $n+1$ and is determined by the system considered.

III. SECOND LAW

There are processes, or motions, that satisfy the First Law but are not observed in nature. The purpose of the Second Law is to incorporate such experimental facts into the model of dynamics. The statement of the Second Law is made using the axiomatic statement provided by the Greek mathematician Carathéodory[4], who presented an axiomatic development of the Second Law of thermodynamics that may be applied to a system of any number of variables. The Second Law may then be stated as follows:

> In the neighborhood (however close) of any equilibrium state of a system of any number of dynamic coordinates, there exist states that cannot be reached by reversible E - conservative ($\bar{d}E = 0$) processes or motions.

Results of the Second Law

One may use the Second Law to show that an integrating factor must exist for the First Law and that this integrating factor must be a function of velocity only. Using the integrating factor the expression for the First Law may be written

$$\bar{d}E = \phi(u)\ f(\sigma)\ d\sigma. \tag{4}$$

Since $f(\sigma)d\sigma$ is an exact differential, the quantity $\frac{1}{\phi(u)}$ is an integrating factor for $\bar{d}E$. It is an extraordinary circumstance that not only does an integrating factor exist for the $\bar{d}E$ of any system, but this integrating factor is a function of speed only and is the same function for all systems. Since the integrating factor is the same form for all systems it does not depend upon the type of force involved and is, therefore, unique.

The universal character of $\phi(u)$ makes it possible to define an absolute speed in the same manner as is done in thermodynamics when defining the absolute temperature. The definition of the absolute speed requires constant speed motions be considered. All Galilean frames of reference will display this process as one of constant speed. Further, if all reference frames are to be of equal status then observers in all Galilean reference frames must share the $\bar{d}E = 0$ constant speed motion equivalently. Furthermore, each observer will have the same value for the absolute speed or else one of the frame will enjoy a privileged nature. Then the absolute speed is unique for all Galilean frames of reference. There is one such speed already known and that speed is the speed of light, c. Therefore, the absolute speed must be the speed of light and the same for all Galilean observers. This is Einstein's postulate. Thus, the first two laws require Einstein's postulate concerning the speed of light.

From Equation (4) we may write

$$\frac{\bar{d}E}{\phi(u)} = f(\sigma)d\sigma. \tag{5}$$

Since σ is an actual function of u and q, the right-hand member is an exact differential, which may be denoted by dS; and

$$dS = \frac{\bar{d}E}{\phi(u)} \tag{6}$$

where S is the mechanical entropy of the system.

IV. GEOMETRY

With the above laws and the definition of the entropy an expression for the generalized Clausius'inequality may be written and used to specify the stability condition

$$\delta U - F_i \Delta q^i - \phi \delta S > 0. \tag{7}$$

which leads to the quadratic form $(ds)^2 = h_{ij} dq^i dq^j$; $j, k = 0, 1, 2, \ldots, n$, where $q^0 = S/F_0$ and

$$h_{jk} = \frac{\partial^2 U}{\partial q^j \partial q^k} \tag{8}$$

The element of arc length may be parameterized using the local time as $ds = cdt$. However, Clausius' Inequality does not lead to a variational principle on time rather it leads to two variational principles, one requiring the minimization of Free Energy and one requiring the maximization of the entropy for isolated systems for which $\bar{d}E = 0$. The differential of the entropy is on the right hand side of this quadratic form so that the form must be solved for this differential expression in order to use the variational principal. When this is done we find that

$$(dq^0)^2 = f(c^2dt^2 + 2h_{0\alpha}cdq^\alpha dt - h_{\alpha\beta}dq^\alpha dq^\beta). \tag{9}$$

Equation (9) displays the relativistic relationship between space and time and that $(dq^0)^2 = f(d\epsilon)^2$. This shows the requirement for two metric spaces coupled by a gauge function, f. Since the Second Law requires that the entropy is a total derivative one may suspect that the entropy space will be an integrable space and this is indeed the case when the Second Law is applied to the metric coefficient. In addition, one finds that the second space, which we might call an energy space because of the tie to the First Law, must be a Weyl space. Therefore, we find that the gauge function acts as a geometrical integrating factor coupling the non-integrable Energy space to the integrable entropy space.

V. QUANTUM MECHANICS

The appearance of Weyl character of the Energy space allows the use of London's work which shows that null trajectories in a Weyl space must be described by the equations of quantum mechanics[4]. In the Dynamic Theory necessity of considering null trajectories comes in a very natural way. For instance, in thermodynamics the desire to consider stable states would cause one to look for isentropic states. This is of course a null trajectory in the entropy space, however, for non-zero gauge functions this condition is also a null trajectory in the energy, Weyl, space. By the Second Law the differential change of entropy can never be negative for an isolated system so that $dq^0 \geq 0$. Therefore, the entropy metric is positive definite. For negative gauge functions the energy space will be negative definite and, therefore, complex. There are an infinite number of null trajectories for a complex space and these are given by the quantum states.

This may be more easily seen by considering the displacement of the element of arc length in the energy space which must take on the Weyl displacement form of

$$d(d\epsilon) = \phi_k dq^k (d\epsilon) \tag{10}$$

where the ϕ_k are the gauge potentials and are the logarithmic derivative of the square root of the gauge function. Then the isentropic condition that the integral of Equation (10) require that $(d\epsilon) = (d\epsilon)_0$ and, therefore,

$$e^{\int \phi_j dq^j} = 1 \text{ so that } \int \phi_j dq^j = 2\pi i N. \tag{11}$$

Equation (12) is the quantum condition that London used to derive Schröedinger's equations of quantum mechanics. Here the quantum condition is required by the isentropic state specification.

VI. FIVE DIMENSIONAL FIELDS

When one first begins to study the thermodynamics of steam systems one writes the First Law as $\bar{d}E = dU + Pdv - F_\alpha dx^\alpha,\ \alpha = 1, 2, 3.$ The right hand side of this statement of the First Law contains five unknown variables. The accepted method of reducing the number of unknowns is to first state that the mass density can always be written as a function of space and time thereby reducing the number of additional independent equations needed to four which are pointed out to be an equation of state and the three mechanical laws of motion from Newtonian physics.

The procedure outlined above for obtaining the equations of the metrics may be used here also and then the dependence or independence of the mass density upon space and time may be determined as the predicted phenomena agree or disagree with experience. This leads one to a five dimensional entropy metric of space-time-mass. Here also one finds the appearance of the two spaces coupled by a gauge function for an isolated system. In this case the gauge function is a function of the same five variables. In the five-dimensional manifolds we have

$$(dq^0)^2 = f(d\epsilon)^2 \tag{12}$$

where

$$\begin{aligned} (dq^0)^2 &= g_{jk} dx^j dx^k \quad j,k = 0,1,2,3,4 \\ (d\epsilon)^2 &= \hat{g}_{jk} dx^j dx^k \quad j,k = 0,1,2,3,4 \end{aligned} \tag{13}$$

with $f = f(t,x,y,z,\gamma)$.

Using standard variational techniques allowing the gauge function to vary one finds that there are eight field equations that are given by

$$\begin{aligned}
\nabla\cdot\bar{B} &= 0 && [a] \\
\tfrac{1}{c}\tfrac{\partial\bar{B}}{\partial t} + \nabla\times\bar{E} &= 0 && [b] \\
\nabla\times\left(\tfrac{\bar{B}}{\mu}\right) - \tfrac{1}{c}\tfrac{\partial(\epsilon V_4)}{\partial t} &= \tfrac{4\pi\bar{J}}{c} - \alpha_0\tfrac{\partial\left(\frac{\bar{V}}{\mu}\right)}{\partial\gamma} && [c] \\
\nabla\cdot(\epsilon\bar{E}) &= 4\pi\rho - \alpha_0\tfrac{\partial(\epsilon V_4)}{\partial\gamma} && [d] \\
\tfrac{\partial\rho}{\partial t} + \nabla\cdot\bar{J} &= -\alpha_0\tfrac{\partial J_4}{\partial\gamma} && [e] \\
\nabla\times\bar{V} &= -\alpha_0\tfrac{\partial\bar{B}}{\partial\gamma} && [f] \\
\nabla V_4 + \tfrac{1}{c}\tfrac{\partial\bar{V}}{\partial t} &= \alpha_0\tfrac{\partial\bar{E}}{\partial\gamma} && [g] \\
\nabla\cdot\left(\tfrac{\bar{V}}{\mu}\right) + \tfrac{1}{c}\tfrac{\partial(\epsilon V_4)}{\partial t} &= -\tfrac{4\pi J_4}{c} && [h]
\end{aligned} \tag{14}$$

For these field equations the force laws are

$$\begin{aligned} K_0 &= (i/c)[\bar{J}\cdot\bar{E} + J_4 V_4] \\ \bar{K} &= \rho\bar{E} + (1/c)(\bar{J}\times\bar{B}) + (J_4/c)\bar{V} \\ K_4 &= \rho V_4 - \tfrac{\bar{J}\cdot\bar{V}}{c} \end{aligned} \tag{15}$$

While the similarity of the E and B fields in the above field and force equations with the well known electric and magnetic fields is obvious. The pressing question concerns the interpretation of the four new V field components. It would be much easier to interpret the physical meaning of the V field if we had the dependence of this new field upon the distance from its source. At the heart of this question is determination of the fields that a particle may possess. In order to see how this question may be addressed, consider the previously stated quantum condition as given by Equation (12). Now suppose we desire to determine the gauge field that a "fundamental" particle may possess. An original notion of a fundamental particle is one that cannot be divided further. The notion we consider here is that we wish to determine what gauge potentials our particle may possess such that its entropy does not change and the gauge potentials remain independent of the path. This requirement means that our particle will retain its gauge potentials over time and while being displaced in space. The procedure to find these potentials is to determine what gauge potentials satisfy the quantum condition independent of the path we choose for the particle.

The first requirement that may be shown is that since we may choose all of the dx to be zero in Equation (12) except a given one of our choosing, then the individual gauge potentials must each be quantatized in unit steps with a different quantum number assigned to each of the gauge potential components. This quantatization is reflected in nature by the unit charges that appear on the various particles. The Dynamic Theory is the only theory known to the author which makes this prediction from the principles of the theory.

The quantatized gauge potentials may be differentiated to obtain the quantatized gauge fields. When the gauge fields are substituted into the field equations, Equations (15), one finds that the gauge potentials depend upon the separation from the site of the particle as $(\frac{1}{r})\ exp(-\frac{\lambda}{r})$. This is a non-singular potential which leads to the following expression for the E and V fields:

$$\begin{aligned} E_r &\propto \left(\frac{1}{r^2}\right)\left(1-\frac{\lambda_N}{r}\right)e^{-\left(\frac{\lambda_N}{r}\right)} \\ V_r &\propto \left(\frac{1}{r^2}\right)\left(1-\frac{\lambda_N}{r}\right)e^{-\left(\frac{\lambda_N}{r}\right)}. \end{aligned} \tag{16}$$

The long range character of the radial component of the E field solidifies the interpretation of the E field as the electric field. The same long range character of the radial component of the V field argues that the V field must be the gravitational field if it is to correspond to experience.

VII. UNIT OF ACTION (QUANTUM MEASUREMENT)

The basis for the quantum measurement or the unit of action rests upon the Poisson Brackets. The quantum Poisson Brackets must correspond to the classical Poisson Brackets and as such must use covariant differentiation. This should come as no surprise since this is what is done to get the correct answer in atomic physics. What may not be so widely known is that when a gauge function is involved the covariant differentiation requires that the unit of action as given by the quantum Poisson Bracket between space and momentum be given by the product of the gauge function and Planck's constant. To see this, consider

$$[x^j, p^j]\Psi = i\hbar\hat{g}^{kl}\left[\delta_{jk} + \binom{j}{s\ l} x^s\right]\Psi \tag{17}$$

so that when the vector curvature becomes negligible the effective unit of action become $f\hbar$ when it is remembered that quantum effects must be determined in the energy, or Weyl, space.

VIII. GRAVITATION

The Dynamic Theory couples gravitation and quantum measurement in explaining the phenomena of the shift of the frequency of light received from distant stars. The effective unit of action for the emission and reception of light has been shown to be[5]

$$\begin{aligned} \hat{\hbar}_e &= \hbar\ exp\left[\frac{W_e(1+bt_e)}{R_e}e^{-\frac{\lambda_e}{R_e}}\right] \\ \hat{\hbar}_r &= \hbar\ exp\left[\frac{W_r(1+bt_r)}{R_r}e^{-\frac{\lambda_r}{R_r}}\right]. \end{aligned} \tag{18}$$

By using the conservation of photon energy, setting the time of emission to zero, setting the time of reception to $L/c, b = -H$, and $W = -GM/c^2$ one obtains the red shift as

$$\frac{\Delta\lambda}{\lambda_e} = exp\left[\left(-\frac{G}{c^2}\right)\left[\frac{M_r e^{\frac{-\lambda_r}{R_r}}}{R_r} - \frac{M_e e^{\frac{-\lambda_e}{R_e}}}{R_e}\right] + \left(\frac{HL}{c}\right)e^{\frac{-\lambda_r}{R_r}}\right] - 1 \tag{19}$$

which has the appropriate limits that match experimental evidence. However, it may be seen that the influence of the exponential term begins to dominate at large distances, L, or for larger gravitational potentials so there should be no surprise in seeing large values of red shifts.

Another gravitational phenomena which involves the gravitational gauge potential is that of the advance of the perihelion of the planetary motions. This may most easily be seen by using the non-singular potential and comparing the orbital frequency with the frequency of small oscillations about the steady circular motion. When this advance is compared to the advance predicted by Einstein's General Theory of Relativity one finds

$$\delta\theta/\delta\theta_{GTR} = (M_1^2 + M_2^2)/(M_1 + M_2)^2 \tag{20}$$

For binary stars where the masses are approximately equal the predicted advance of the Dynamic Theory is only half that of General Relativity as experimental data shows. However, when either body becomes large compared to the other the prediction of the Dynamic Theory goes over to that of General Relativity.

IX. CONCLUSIONS

The thermodynamic basis of the Dynamic Theory requires us to use two spaces coupled by a gauge function which is a five-dimensional function which depends upon space, time, and mass. The coupling of the entropy and the energy spaces by the gauge function requires that, though there is only one null trajectory in the integrable entropy space, it is necessarily accompanied by an infinite number of null trajectories in the non-integrable energy space when a negative gauge potential is used. Therefore, we must conclude that quantum mechanics is not more basic than a geometric theory, but is part of it by being those null trajectories necessary to describe stable, constant, entropy states.

Quantum measurement as given by the uncertainty principle is controlled by the gauge function through the gauge potentials. This was displayed by presenting the manner in which the gravitational gauge potential influenced the unit of action for the atoms emitting light from a distant star and the unit of action here on earth with which the frequency is compared.

If locality is determined by comparing possible velocities to the speed of light then the Dynamic Theory gives the following five-dimensional limiting time rate of change of the entropy

$$\frac{dq^0}{dt} = \sqrt{c^2 - v^2 - (g_{44}/a_0^2)\dot{\gamma}^2}. \tag{21}$$

The limiting aspect of Equation (22) occurs when the time rate of change in the entropy goes to zero. In free space, the limiting velocity must be the speed of light. However, this is not true in general and may exceed the speed of light should the metric coefficient of the mass density be negative.

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